A Statistical Analysis of Prize Money, Tournaments, and Players (1998–2025)

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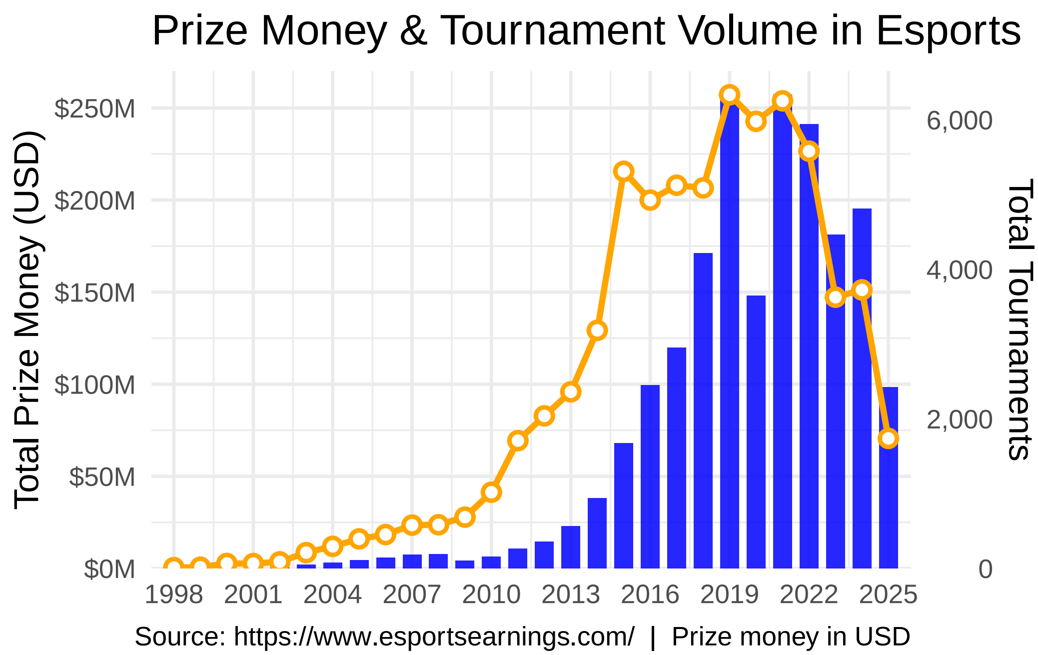
1 Introduction

E-sports has evolved from a small group of enthusiasts' entertainment into a global industry over the past three decades. At the same time, the number of professional players, the frequency of the competition, and the number of players are continuously increasing, which leads to esports becoming a formal exercise. However, we still lack quantitative research on the long-term growth model of e-sports. For instance, over the past nearly three decades, how has the total prize money of e-sports changed over time? Are there significant differences in the distribution of individual player incomes between different years? And is there a clear correlation between the growth of active players and the total prize money of. In order to answer these questions, this report records the data of e-sports from 1998 to 2025, including total award money, total number of tournaments, mean and median tournament award pool, and mean and median earnings per player. events? This report's analysis part mainly includes three parts: the trend of the award pool of the tournaments over the years, the difference between the mean and the median, and the relationship between the number of players and the award pool of the tournament.

2 Data Analysis

2.1 The Relationship between the Total Prize Pool and The Number of Tournaments

Figure 1: Annual esports prize money (bars, left axis) and total tournaments held (line, right axis) from 1998 to 2025. Prize money values have been scaled to millions of USD for readability.



2.1.1 Analysis

This plot helps show how the overall scale of esports changed from 1998 to 2025 by comparing two measures at the same time: **total prize money** and **total tournaments**. The blue bars show how much prize money was awarded each year, while the orange line shows how many tournaments were held. Looking at both together makes it easier to see whether esports growth came from having **more events**, **larger prize pools**, or both.

There are three broad phases in the graph. **First**, from the late 1990s through about the early 2010s, both prize money and tournament counts were relatively low, though they gradually increased over time. **Second**, from about 2013 through 2019, esports expanded very quickly. Tournament counts rose sharply, and total prize money also rose at a fast rate. **Third**, the late 2010s to early 2020s appear to be the high point, with prize money peaking around 2021 and tournament counts remaining very high from about 2016 through 2022.

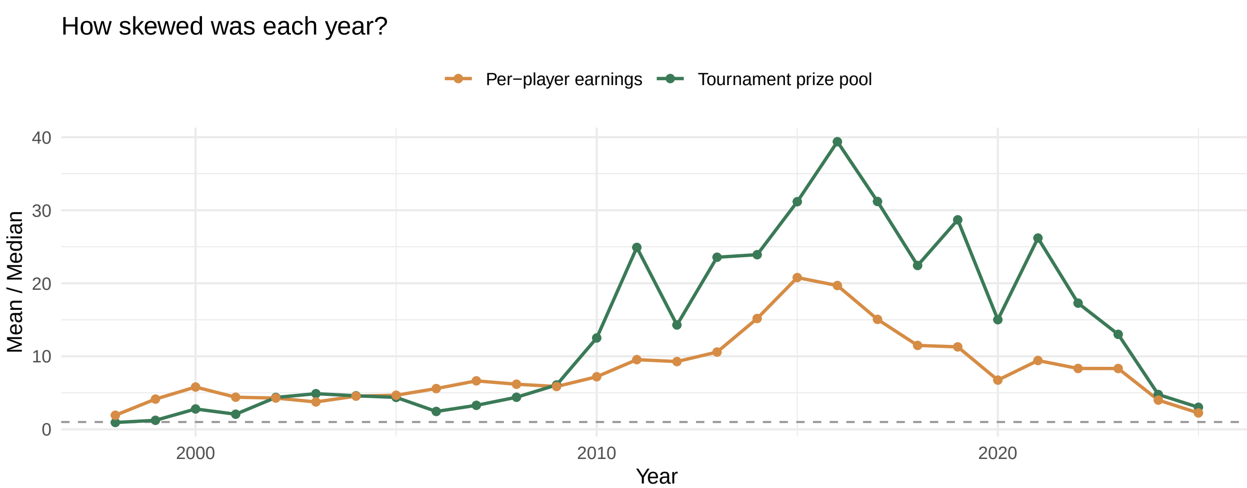
Since 2022, both metrics have trended downward, with 2025 data showing the lowest tournament count since 2013. This may reflect broader economic headwinds affecting esports investment, or a structural shift in how competitive gaming is organized and monetized.

A key insight from this graph is that **prize money and tournament volume do not always move at exactly the same rate**. For example, tournament counts remain high for several years even when prize money fluctuates more strongly. That suggests the esports ecosystem can stay active even when the money side changes, and it may also show that some years had fewer or more concentrated high-prize events.

2.2 Mean VS. Medium(JunBoZhao)

Figure 2: Mean and median over time, on a log scale. Left: prize pool per tournament. Right: earnings per active player.

Figure 3: Ratio of mean to median over time. A value of 1 (dashed line) would mean a perfectly balanced distribution; everything above it is right-skew.



2.2.1 Analysis

We wanted to compare the mean and median here because they pick up on different things. The mean gets pulled around by outliers, one massive tournament like The International will push it way up even if most events that year were small. The median does not care about that at all; it just shows what the middle tournament looks like. So when these two numbers are far apart, it usually means the distribution is really skewed.

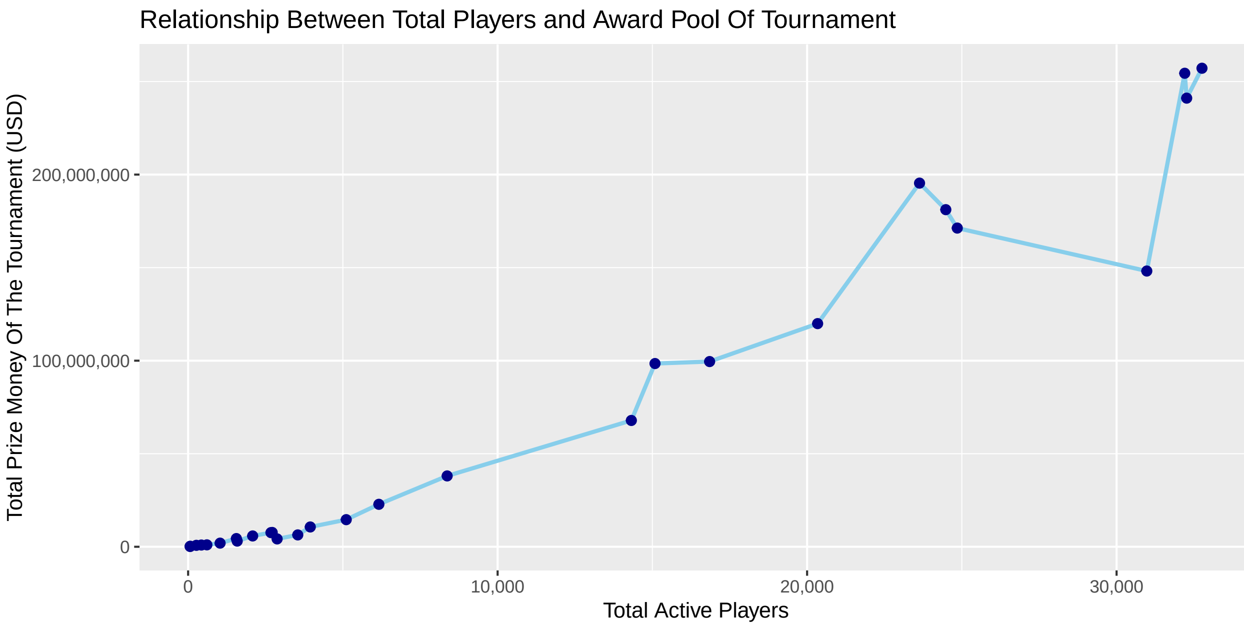
First, looking at Figure 2, we had to use a log scale because the range is kind of ridiculous. In 2010 the median tournament was paying around $250, but the mean was already over $6,000. On a normal axis you would not even be able to see the median line. From about 2010 to 2021, the two lines on the tournament side barely look related: the median stays somewhere in the low hundreds to low thousands while the mean is up in the tens of thousands. We plotted the actual ratio in Figure 3 to get a clearer picture, and for most of that stretch it is somewhere between 12x and 29x. Basically, for every big event like LoL Worlds there were a ton of small local tournaments that only paid out a couple hundred dollars, and those kept dragging the median down.

Second, on the player earnings side, the gap is there too but not as extreme. The mean is usually around 3 to 7 times the median, with a peak at about 15x in 2014. So a handful of top players were making a lot, but most people competing were earning way less than the average would suggest.

What caught our attention was the shift after 2022. The median tournament prize pool went from roughly $1,500 in 2021 up to over $18,000 by 2025, and the mean-to-median ratio dropped from around 26 all the way to about 3. At first that looks like a good thing, like the money is spreading out more evenly. But when we checked the total number of tournaments, the count went from about 6,300 in 2019 down to around 1,700 in 2025. So what actually happened is a lot of the smaller grassroots events stopped running. The tournaments that are left tend to be bigger and better funded, which is why the median jumped up and closed the gap with the mean. It is less that the distribution got more equal and more that the bottom end just went away.

2.3 The Relationship Between The Number Of Players And The Award Pool Of The Tournament

Figure 4: Relationship Between Total Players and Award Pool Of Tournament



2.3.1 Analysis

As the Figure 4 shown, overall, there is a strong positive correlation.That is, the more active players there are, the higher the total prize money for the event. This is intuitive, as an increase in participants usually means an expansion of the market size, which attracts more sponsors, viewers and capital inflows, thereby raising the prize pool. Moreover, this line is not a uniform straight line, it can be split into three parts. First is player number from 0 to 10000, in this case the growth of bonuses has been very slow, which is in line with the early stage of the e-sports industry. At that time, the player base size was small and the commercialization level was low. Second, is player number from 10001 to 22000, in this case the slope has significantly steepened, and the bonus amount has rapidly increased from several hundred million dollars to nearly 200 million dollars. This indicates that within this range, the increase in bonus for each additional active player is much higher than in the early stages. Finally, is player number greater than 22001. An interesting phenomenon emerges here: the number of players decreases first and then rises sharply. This indicates that the relationship between the number of players and the prize money is no longer strictly linear. In some years, there are more players but the prize money is actually lower. Although there is a strong positive correlation shown in the graph, it cannot be simply concluded that "more players lead to more bonuses". Because supply and demand are mutually restrictive: as the bonus pool expands, it attracts more players to join, while an increase in the player base in turn attracts sponsors to make further investments, forming a positive feedback loop.

3 Summary

Analysis of the graph narrative

4 Code Appendix

```r
library(tidyverse)
library(ggplot2)
library(scales)

# Section 1: The Relationship between the Total Prize Pool and The Number of Tournaments
# Author: Seokyoung Kim

df <- read_csv("Esports_Stats.csv") |>
  rename(
    year              = year,
    prize_money       = `Total Prize Money`,
    total_tournaments = `Total Tournaments`
  ) |>
  select(year, prize_money, total_tournaments)

scale_factor <- max(df$prize_money) / max(df$total_tournaments)

col_prize  <- "blue"   # blue  - prize money
col_tourn  <- "orange"   # orange - tournament

p <- ggplot(df, aes(x = year)) +

  geom_col(
    aes(y = prize_money),
    fill  = col_prize,
    alpha = 0.85,
    width = 0.7
  ) +
  geom_line(
    aes(y = total_tournaments * scale_factor),
    color     = col_tourn,
    linewidth = 1.1
  ) +
  geom_point(
    aes(y = total_tournaments * scale_factor),
    color = col_tourn,
    size  = 2.2,
    shape = 21,
    fill  = "white",
    stroke = 1.2
  ) +

  # Dual y-axes
  scale_y_continuous(
```

```r
    name   = "Total Prize Money (USD)",
    labels = function(x) paste0("$", x / 1e6, "M"),
    expand = expansion(mult = c(0, 0.05)),
    sec.axis = sec_axis(
      transform = ~ . / scale_factor,
      name      = "Total Tournaments",
      labels    = function(x) formatC(x, format = "d", big.mark = ",")
    )
  ) +
  scale_x_continuous(
    breaks = seq(1998, 2025, by = 3),
    expand = expansion(add = 0.5)
  ) +

  theme_minimal(base_size = 13) +

  labs(
    title    = "Prize Money & Tournament Volume in Esports",
    x        = NULL,
    caption  = "Source: Esports_Stats.csv  |  Prize money in USD"
  )
p

# Section 2: Mean vs. Median
# Author: Junbo Zhao

# Figure 1
dataTable <- read.csv("Esports_Stats.csv", check.names = FALSE)

mvmLong <- dataTable |>
  select(year,
         `Mean Tournament Prize Pool`,
         `Median Tournament Prize Pool`,
         `Mean Earnings/Player`,
         `Median Earnings/Player`) |>
  pivot_longer(-year, names_to = "metric", values_to = "value") |>
  mutate(
    panel  = ifelse(grepl("Tournament", metric),
                    "Per tournament prize pool",
                    "Per player earnings"),
    centerType = ifelse(grepl("^Mean", metric), "Mean", "Median")
  )

ggplot(mvmLong, aes(x = year, y = value, color = centerType)) +
  geom_line(linewidth = 0.8) +
  geom_point(size = 1.6) +
  facet_wrap(~ panel, scales = "free_y") +
```

```r
  scale_y_log10(labels = dollar) +
  scale_color_manual(values = c("Mean" = "#c0392b", "Median" = "#2c3e50")) +
  labs(
    title = "Where the average meets the typical",
    x = "Year", y = "USD (log scale)", color = NULL
  ) +
  theme_minimal(base_size = 11) +
  theme(legend.position = "top",
        strip.text = element_text(face = "bold"))

# Figure 2
dataTable <- read.csv("Esports_Stats.csv", check.names = FALSE)

ratioLong <- dataTable |>
  transmute(
    year,
    `Tournament prize pool` =
      `Mean Tournament Prize Pool` / `Median Tournament Prize Pool`,
    `Per-player earnings`   =
      `Mean Earnings/Player`     / `Median Earnings/Player`
  ) |>
  pivot_longer(-year, names_to = "metric", values_to = "ratio")

ggplot(ratioLong, aes(x = year, y = ratio, color = metric)) +
  geom_hline(yintercept = 1, linetype = "dashed", color = "grey60") +
  geom_line(linewidth = 0.8) +
  geom_point(size = 1.6) +
  scale_color_manual(values = c("Tournament prize pool" = "#3b7a57",
                                "Per-player earnings"   = "#d68c45")) +
  labs(
    title = "How skewed was each year?",
    x = "Year", y = "Mean / Median", color = NULL
  ) +
  theme_minimal(base_size = 11) +
  theme(legend.position = "top")

# Section 3: The Relationship Between The Number Of Players And The Award Pool Of The Tournamen
# Author: JiXian Li

dataTable <- read.csv("Esports_Stats.csv", check.names = FALSE)
ggplot(dataTable, aes(x = `Total Active Players`, y = `Total Prize Money`)) +
  geom_line(color = "skyblue", linewidth = 1) +
  geom_point(color = "darkblue", size = 2) +
  scale_x_continuous(labels = comma) +
  scale_y_continuous(labels = comma) +
  labs(
    title = "Relationship Between Total Players and Award Pool Of Tournament",
```

```
    x = "Total Active Players",
    y = "Total Prize Money Of The Tournament (USD)"
  )
```